
THE EFFECTS OF COMBINING ELASTIC AND FREE WEIGHT RESISTANCE ON STRENGTH AND POWER IN ATHLETES

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ABSTRACT

This study was undertaken to determine whether combined elastic and free weight resistance (CR) provides different strength and power adaptations than free weight resistance (FWR) training alone. Forty-four young (age 20 ± 1 years), resistance-trained (4 ± 2 years' experience) subjects were recruited from men's basketball and wrestling teams and women's basketball and hockey teams at Cornell University. Subjects were stratified according to team, then randomly assigned to the control (C; $n = 21$) or experimental group (E; $n = 23$). Before and after 7 weeks of resistance training, subjects were tested for lean body mass, 1 repetition maximum back squat and bench press, and peak and average power. Both C and E groups performed identical workouts except that E used CR (i.e., elastic resistance) for the back squat and bench press, whereas the C group used FWR alone. CR was performed using an elastic bungee cord attached to a standard barbell loaded with plates. Elastic tension was accounted for in an attempt to equalize the total work done by each group. Statistical analyses revealed significant ($P < 0.05$) between-group differences after training. Compared with C, improvement for E was nearly three times greater for back squat (16.47 ± 5.67 vs. 6.84 ± 4.42 kg increase), two times greater for bench press (6.68 ± 3.41 vs. 3.34 ± 2.67 kg increase), and nearly three times greater for average power (68.55 ± 84.35 vs. 23.66 ± 40.56 watt increase). Training with CR may be better than FWR alone for developing lower and upper body strength, and lower body power in resistance-trained individuals. Long-term effects are unclear, but CR training makes a meaningful contribution in the short term to performance adaptations of experienced athletes.

KEY WORDS variable resistance, physical training, bungee cords, rubber bands

INTRODUCTION

The mechanics involved in the use of free weights is, at times, problematic for achieving maximal effort. Because of the body's lever systems, the maximal weight that can be used during an exercise is limited by a phenomenon commonly known as the sticking point. The sticking point creates a most disadvantaged position in the range of motion (i.e., near the bottom of a squat and bench press), thereby effectively limiting the maximal external resistance (6). It also negatively affects both average force and acceleration achieved during a free weight resistance (FWR) exercise. If this sticking point could be minimized, a more uniformly distributed external resistance would allow for greater loading at positions of greater leverage. As a result, greater average muscle tension could be achieved throughout the range of movement, and greater strength gains might be realized.

Equipment manufacturers have long recognized the biomechanical disadvantages to FWR training and have developed numerous alternatives to create a variable nature of external resistance. However, machines typically are expensive and may not be any more effective than FWR training at strength development (19,23). Recently, combination weight training techniques have increased in popularity. One such method combines FWR with large elastic bands in an attempt to merge the benefits of variable resistance with those of free weights in a technique that is inexpensive and readily available. Employing a combination of elastic resistance (ER) and FWR (i.e., combined resistance [CR]) may be useful for minimizing the disadvantages of FWR, namely the sticking point, and for developing strength and power in athletes.

With CR, the external resistance during an exercise should theoretically be more evenly distributed over the full range of motion than with FWR exercise. CR should decrease the need for dramatic deceleration resulting in an extended period of acceleration during the movement (18). Greater acceleration translates into greater average force development throughout the whole range of motion than would have otherwise been encountered with either ER or FWR exercises alone. Greater muscle tension development during each repetition, and each set performed, should provide for

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greater neuromuscular stimulus than experienced with traditional FWR exercises. Therefore, in principle, using CR should provide greater opportunity for muscular adaptation and strength development than FWR training alone (20,25). Despite the rationale for applying CR in strength training, there is very little empirical research of this potentially effective technique.

The purpose of this study was to examine the effect of using CR during off-season training in a variety of Division I-A athletes. It was hypothesized that CR would provide an additional stimulus for neuromuscular adaptation than traditional FWR exercise, leading to a greater improvement in muscle performance after 7 weeks of training.

METHODS

Experimental Approach to the Problem

This study used a stratified sampling of 44 trained Division I-A athletes (non-scholarship) recruited from men's basketball and wrestling teams and women's basketball and hockey teams at Cornell University to determine whether CR enhances strength and power adaptations compared with FWR alone. Treatment and control groups were randomly assigned in a balanced fashion within teams so that 22 men and 22 women began the program with equal numbers of each sex per group. A 7-week macrocycle was selected as the training program duration because this practically and realistically fits within the constraints of the spring academic calendar for athletes who had recently completed competing in winter sports. Strength, power, and body composition were assessed using a pre-post test design with the treatment group using CR training for 7 weeks, while the control group trained using FWR only. Elastic resistance is a technique that has gained in popularity in recent years yet has not been empirically tested. The value of using elastic bands needs to be better understood to make more informed decisions when coaching and training athletes.

Subjects

Each subject completed a questionnaire describing past experience in sport and resistance training, which documented no substantial between-group differences with respect to training history (treatment: 3.7 years; control: 3.6 years). Subjects had no previous experience with CR training. All subjects were experienced with at least 2 years of consecutive weight training in an organized and periodized program. This project was conducted as part of a voluntary after-season conditioning program with all teams 2–6 weeks post-competition and each athlete beginning a new training year cycle. Before participating in this study, all subjects read and signed an informed consent previously approved by the institutional research review board at Cornell University.

Forty-four young (aged 20 ± 1 years) athletes (22 men and 22 women) completed the study, with 5 of the initial 44 subjects failing to complete all data collection because of injuries incurred outside the training program. In the end, all

subjects completed the bench press and body composition assessments; however, only 39 were tested post training for back squat and power. Adherence to the training program was extremely high, yielding an attendance rate greater than 99% (among all subjects, only 4 of 968 sessions were missed).

Procedure

The subject sample was stratified by team, and each team was divided into equal subsets of treatment and control using partial random assignment. A pre-post testing design was employed with treatment subjects using CR training techniques for 7 weeks, while the control group trained in a similar fashion using FWR only. Assessments were performed during 2 days and completed in an identical manner before and after training. Testing examined muscular strength, power, and body composition.

Strength Assessments. Bench press and back squat 1 repetition maximum (1RM) were estimated from a 1–3 RM effort using the equation described by Wathan (27). Each athlete was given three to six attempts with progressively increasing weight to achieve a 1–3 RM with 3–5 minutes' rest between attempts. This technique allows for adequate warm-up without inducing excessive fatigue (7). Although 1RM testing is preferable to 1–3 RM testing for accuracy, we were limited by University policy for strength testing that required 1–3 RM testing. While testing, subjects wore support belts but were not permitted to wear knee wraps or squat suits. For the bench press, subjects were required to keep their hips on the bench, and the bar had to travel from touching the chest to full arm extension for a complete repetition. No chest bouncing was permitted, and their grip was between the outer rings and inner edge of the bar knurling on a standard Olympic bar (no close grips or wide grips were allowed). The squat was performed to a depth with the hip joint being parallel with the knee joint using a stance slightly wider than shoulder width. Repetitions not done to parallel were not counted, and wide power lifting or narrow Olympic squat stances were not allowed. Experienced and certified strength and conditioning coaches supervised all testing.

Power Assessments. A countermovement vertical jump (CVJ) off two feet and with full arm motion was employed to predict peak and average power (8,11). Standing reach measures were subtracted from Vertec (Sports Imports, Columbus, OH)-determined jump height to calculate CVJ.

Body Composition Assessments. Pre- and post-training measures were made at approximately the same time of day using John Bull skin fold calipers (British Indicators, Ltd., London, UK). A generalized seven-site formula for college-aged men (12) and women (13) was used to estimate lean body mass (LBM). All testing was performed at the same time of day, and the same individual performed all skinfold testing. The test-retest reliability was $r = 0.99$ for the skinfold tester.

The testing schedule and order were the same before and after training. On the first day, the CVJ was completed and followed by the back squat with at least 30 minutes' rest between the two tests. The second day of testing was at least 48 hours after the first day and started with body composition assessment followed by the bench press.

Training Program. Upper body (bench press, one-armed row, triceps extension, lat pull-down, and standing external shoulder rotations [Cuban press]) and lower body (back squat, power shrug, Romanian deadlift [semi-stiff leg], walking lunge, dumbbell curl, and shoulder press combo) exercises were performed on alternate days three times each week. Sets (three to six), repetitions (2–10), and intensity (72–98% 1RM) were periodized in a wavelike progression during 7 weeks of training. Approximately 2–3 minutes' rest was given between sets. If fatigue occurred near the end of a set, subjects were allowed to rest 5–10 seconds between repetitions so that they completed every repetition of every set assigned without assistance from a spotter.

CR was only used for the bench press and back squat exercises. Elastic bands (BNS Bungee System, Power-Up USA, Milwaukee, WI) were attached to the bottom of the power rack and around the bar (see Figures 1 and 2). Elastic tension was accounted for in an attempt to equalize the total work done by each group so that the CR group used approximately 80% of the free (plate) weight used by the FWR group for each exercise. Three different size bands were used to achieve the desired average elastic tension of 20% of the subject's 1RM for the bench press and back squat. Because resistance in CR varies by position within the range of motion, the term "average resistance" was used to determine the load for each exercise. Elastic tension provided by the bands was determined using a 100-lb archery spring scale (Hanson Archery, Shubuta, MI). A regression formula



Figure 1. The bench press set up with elastic resistance in place. Carabiners are used to attach the elastic band to d-rings fixed to the rack at the bottom and the bar at the top.



Figure 2. The back squat rack set up with elastic resistance in place. Carabiners are used to attach the elastic band to d-rings fixed to the rack at the bottom and the bar at the top.

was obtained using the elastic tension at various lengths of displacement. The total bar displacement of each subject in the CR group was measured for both the back squat and bench press. This allowed for individual adjustment of the elastic tension relative to subjects' individual bar displacement in both the back squat and bench press. Resistive load was equalized between the CR and control (FWR) groups by determining the average elastic tension provided by the band and subtracting that weight from the bar weight for the CR group. Actual average elastic tension was $20 \pm 2.6\%$ and $19 \pm 3.3\%$ of the subjects' 1RM for the bench press and back squat, respectively. The amount of band tension, once individualized per subject, remained constant for all training sets during the training period. When adjustments in overall resistance were needed, only the free weight component of CR was varied.

Statistical Analyses

A 2×2 analysis of variance (group $\times$ time) with repeated measures on the time factor was used to examine data for bench press 1RM, back squat 1RM, peak power, average power, and LBM. Significance was set at $P \leq 0.05$ with significant interactions further examined using Tukey's post hoc analyses. In planning the experiment, a statistical power >0.8 was desired. Considering the 1RM tests as the primary dependent variables and by estimating $ES = 15$ kg; $SD = 25$ kg; and $\alpha = 0.05$ for single-sided results in pairwise analysis, our sample size ($n = 22$ per group) yielded a calculated power of 0.88.

RESULTS

As a result of non-training-related injuries or failure to adhere, 5 of 44 original subjects were excluded from some data analyses. In support of our hypothesis, the primary finding reported below is that elastic resistance was effective at

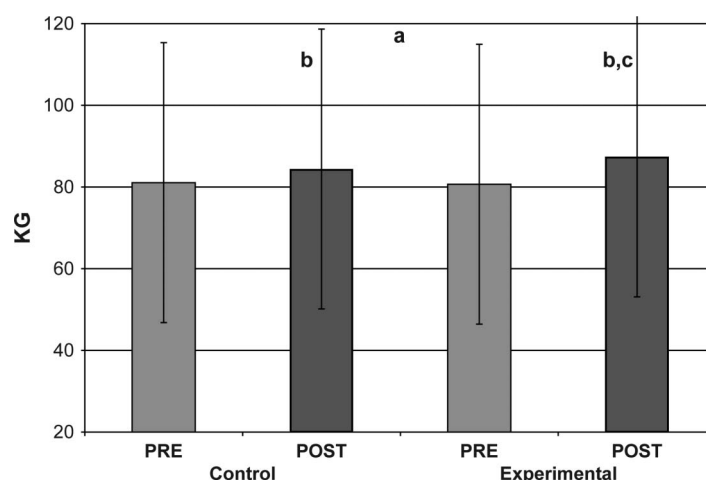


Figure 3. Bench press 1 repetition maximum before and after training. Mean $\pm$ SD for each group with significant differences ($P < 0.05$) denoted by (a) interaction, (b) time within group, and (c) between groups post training.

supplementing weight training in these athletes. Training with CR yielded improved results in bench press and back squat when compared with traditional training with FWR.

Bench Press

A significant interaction ($P < 0.01$) followed by post hoc testing revealed that groups were not different for bench press 1RM before training, but CR was significantly greater than FWR after training (Figure 3). As expected, both groups significantly improved bench press performance with training ($P < 0.05$), but CR experienced a greater

improvement than the group trained with FWR only. The FWR group experienced a 4.0% improvement (from 81.07 ± 33.37 to 84.41 ± 33.37 kg), whereas the CR group improved by 8.0% (from 80.69 ± 35.34 to 87.37 ± 35.52 kg).

Back Squat

As demonstrated by the back squat 1RM, lower body strength followed a pattern similar to the bench press results (Figure 4). Tukey's follow-up to a significant interaction ($P < 0.01$) revealed no difference before training but a significance difference between the CR and FWR groups post training

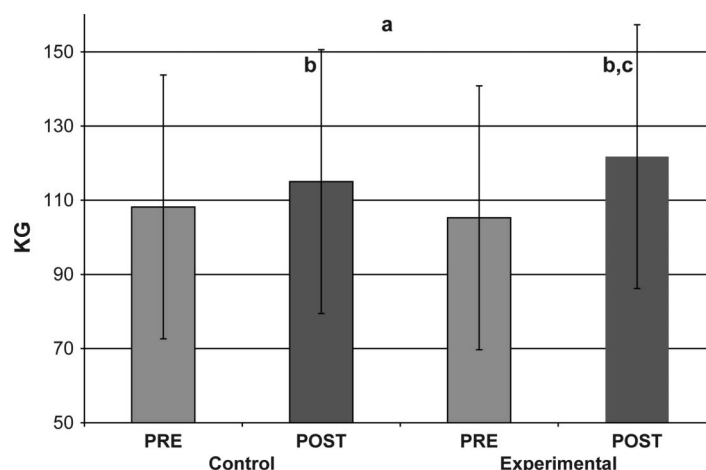


Figure 4. Back squat 1 repetition maximum before and after training. Mean $\pm$ SD for each group with significant differences ($P < 0.05$) denoted by (a) interaction, (b) time within group, and (c) between groups post training.

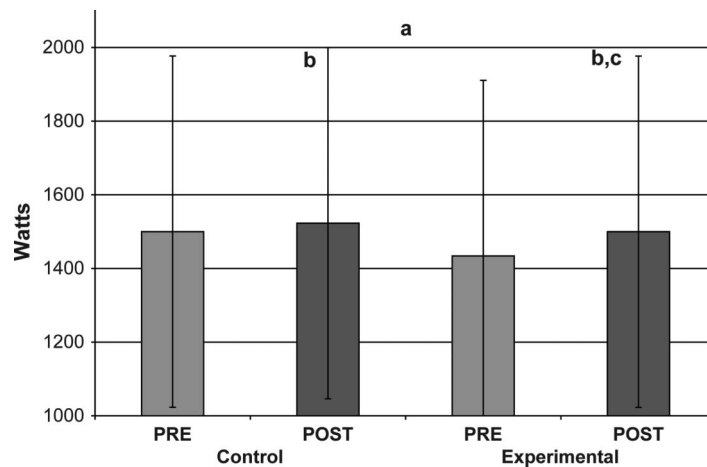


Figure 5. Average power before and after training. Mean $\pm$ SD for each group with significant differences ($P < 0.05$) denoted by (a) interaction, (b) time within group, and (c) between groups post training.

($P < 0.05$). Both groups improved back squat 1 RM during the 7-week training period, but the CR group (16%: from 105.28 ± 33.70 to 121.75 ± 35.70 kg) improved more than twice as much as the FWR group (6%: from 108.19 ± 35.61 to 115.03 ± 37.29 kg).

Power

Average power as calculated from the CVJ revealed a significant interaction ($P < 0.05$); however, peak power displayed only a significant time effect ($P < 0.01$). Power defined by either measure was similar before training and improved over time ($P < 0.05$) in both groups. Post-training

testing showed that average power improvement was significantly greater ($P < 0.05$) in the CR group (from 1434.03 ± 438.15 to 1499.85 ± 471.06 W) than in the FWR group (from 1500.02 ± 500.82 to 1523.68 ± 497.27 W) (Figure 5). The change in peak power was not significantly different between the groups, although this interaction neared the critical probability value ($P = 0.087$) (Figure 6).

Lean Body Mass

Body composition analysis revealed a small but significant increase ($P < 0.05$) in LBM during training for both groups. LBM increased from 66.5 ± 15.5 to 66.8 ± 15.1 kg in the

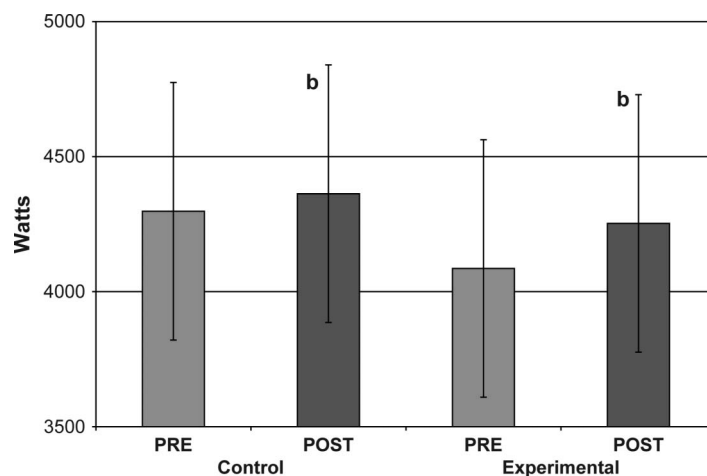


Figure 6. Peak power before and after training. Mean $\pm$ SD for each group with significant differences ($P < 0.05$) denoted (b) time within group.

FWR group and from 65.8 ± 12.1 to 66.5 ± 12.0 kg in the CR group. Increases in LBM are common in well-planned weight training studies. However, in this study, there were no differences in LBM between the groups before or after the training program. In other words, we have no evidence that muscle mass differences accounted for the observed performance differences between the groups.

DISCUSSION

In support of our hypothesis, the main finding of this study is that using heavy elastic bands to alter resistance patterns during weight training allowed for greater gains in strength and power measures during a 7-week training period compared with traditional weight training. This result was found in an already well-trained group of National Collegiate Athletic Association Division I-A athletes. The mechanism accounting for the better improvement seen with CR is unclear; however, these between-group differences are likely related to neural and/or muscular adaptations.

It is well understood that neuromuscular adaptations are specific to the nature of the load imposed on the system during chronic training (3,17,22). It is conceivable that different load distribution characteristics during CR weight lifting altered muscle recruitment patterns. It was observed that subjects in the CR group were always able to complete their prescribed sets, whereas some in the control group were forced to pause (5–10 seconds) between some repetitions to complete sets. Contractions with CR should have kept the muscle working closer to maximal capacity throughout the range of motion but reduced the biomechanical disadvantages throughout the range of motion, which may have altered recruitment patterns. Specifically, contractions with a less acute sticking point or protracted pause may have invoked greater type IIx muscle fiber recruitment and therefore greater adaptation in these fibers. It may be that contractions with CR were done nearer to muscle failure through the range of motion. In fact, Wallace et al. (26) have recently demonstrated that adding elastic resistance to FWR back squats yields greater peak force and peak power during high-intensity efforts (near 85% 1RM), like those used in the present study. Such differences in the nature of contraction during each repetition may have led to differential adaptations, accounting for better performance during post-training testing for the CR group. Results from this study do little to further this speculation because both groups equally improved LBM to a small extent. Even this quantitative finding is subject to limitations inherent in using skinfolds for body composition analysis. Differences in the quality of muscle were not examined, as this was a fairly large-scale training study without access to muscle fiber typing. Speculation about CR causing distinct fiber recruitment or adaptation patterns with training need to be investigated in future studies using more sophisticated imaging techniques.

With traditional FWR exercise, the barbell is accelerated during muscle shortening until the so-called sticking point.

Once this point of minimal leverage is overcome, the force needed to complete the contraction becomes submaximal and the barbell naturally decelerates (6). Behm and Sale (2) pointed out that effort to accelerate a load is a key component to strength gains. With CR, the bar decelerates less through the whole range motion because of increasing elastic resistance that does not exist with free weights alone. Compared with the FWR group, the CR group experienced approximately 10% less resistance at the bottom of a movement and 10% more resistance near the top of the movement. It is likely that acceleration remained constant for a longer period of time during a repetition using CR than FWR, and deceleration was less with CR than FWR. Assuming that acceleration was greater for CR, then different fiber recruitment may occur between the groups during each repetition. Such differences might contribute to a neuromuscular adaptation differential during a training period that may account for the results of the present study.

The eccentric portion of an exercise is an often overlooked, but it is a potentially critical aspect of each repetition. The addition of elastic recoil during the eccentric contraction with CR may pose a different challenge to the neuromuscular system during each repetition. Lowering of the weight during CR may require more fiber recruitment and impose effort at a greater percentage of capacity than without CR. Häkkinen et al. (9) demonstrated increased electromyography (EMG) activity with increasingly greater eccentric contractions. EMG data were not collected during the present study. However, Cronin et al. (4) concluded that CR caused significantly greater eccentric EMG activity when elastic bands were added to a jump squat machine than when the jump squat machine was used without the bands. Greater muscle fiber recruitment and stimulation during the eccentric portion of each repetition may bring about greater neuromuscular adaptations and/or type IIx fiber recruitment with CR than with free weights alone. This speculation is consistent with findings of Nardone et al. (15), who showed preferential recruitment of large motor units during high force eccentric contractions. Raue et al. (16) also demonstrated some differences in adaptation of the myosin heavy chain hybrids when comparing concentric and eccentric contractions. It is possible that, with CR, the nature of contraction is altered during both the concentric and eccentric portions of the effort. Such differences in contraction could account for the long-term muscle performance adaptations seen with CR training.

Previous reports of enhanced performance results when training using CR come from a variety of sources (4,18,21,25). Only the work of Cronin et al. (4) appears in peer-reviewed publication, and they found better lunge performance after 10 weeks of CR machine squats jumping compared with simple machine squat jumping. The anecdotal reports of Simmons (21) and Tate (25) are of interest because they claimed dramatic improvements for elite power lifters in back squat and bench press after using CR. The present study used

protocols similar to those in the anecdotal reports of Simmons (21) and Tate (25). The subjects in the present study were all collegiate athletes in excellent condition. They had ended their competitive season and were beginning post-season training programs. It is considerably more difficult to make performance gains in well-trained individuals than in novice performers (1,10,14). This is the first rigorous experimental study to demonstrate the usefulness of employing CR with trained athletes. The most important observation for CR may be that it can assist in the development of strength and power performance of already well-trained athletes.

PRACTICAL APPLICATIONS

Ebben and Jensen (5) suggested that CR may be a difficult technique to implement with large numbers of athletes. In contrast to that, we found that the elastic bungee system was quite easy to set up. There were no problems posed by installation or utilization for bench press or squat exercises. When the racks were used for training without CR, it was simply a matter of disconnecting the band from the barbell. CR proved to be a relatively inexpensive and easy to use adjunct to free weight training.

Variety is considered key to a well-planned and successful resistive training program. Periodization and training variations throughout the year are needed to bring athletes to peak performance during their competitive season (24). The application of CR may become another tool to provide variety as athletes move through their yearly training cycles. It may be that short cycles of CR can help an athlete to overcome the plateaus that occur in strength and power performance. The present study only studied 7 weeks of training and no conjecture can be made about how CR would fare as a training tool in a longer study. Future studies also need to examine the use of CR in a variety of potential applications throughout the microcycles of a training year for a variety of athletes. It may also be of interest to vary the degree of elastic tension on the bar, as a low percentage of elastic tension (e.g., 10% of 1RM) may be more useful to strength gains and a high percentage of elastic tension (e.g., 30% of 1RM) may be better for enhancing explosive power production. CR has the potential to be a useful tool in the implementation of a successful training program. Determining exactly how and when to use this accessible training aid is important to making precise recommendations for the application of CR.

In conclusion, the present study demonstrated that CR is an effective adjunct to resistive training. Both upper and lower body strength and average power production (lower body) can be enhanced when CR is added to free weight training. These performance enhancements may be related to the altered contractile characteristics associated with raising and lowering a load when CR is employed. Future research will help to best delineate how to use CR during the course of an athlete's training program.

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